Maximum Building II

Time limit: 1.00 s **Memory limit:** 512 MB

You are given a map of a forest where some squares are empty and some squares have trees.

You want to place a rectangular building in the forest so that no trees need to be cut down. For each building size, your task is to calculate the number of ways you can do this.

Input

The first input line contains integers n and m: the size of the forest.

After this, the forest is described. Each square is empty (`.`) or has trees (`*`).

Output

Print n lines each containing m integers.

Constraints

- $1 \le n, m \le 1000$

Example

Input

```
4 7
...*.*.
.*.....
.......
......*
```

Output

```
24 17 13 9 6 3 1
16 9 7 5 3 1 0
9 3 2 1 0 0 0
3 0 0 0 0 0 0
```

Explanation: For example, there are 5 possible places for a building of size 2×4.